Qualitative Analysis Cheat Sheet

Who This Is For

- First-time analysts doing survey thematic analysis.
- Teams using the local pipeline in this project.

1) Quick Start

1. Put your file in `data/` (CSV, TSV, XLSX, or JSON).
2. Edit `config/config.example.yaml` or copy it to `config/my_project.yaml`.
3. Set:
- `input.file_path`
- `input.text_column`
- `input.question_column` (recommended)
4. Run:
- VS Code task: `Survey NLP: Run (pick config)`, or
- `python analyze_survey.py --config config/my_project.yaml`
5. Open outputs in `output/`.

2) Analysis Modes

General Qualitative Analysis

- Global clustering across all responses.
- Best when you want overall themes.
- Key outputs:
- `cluster_summary.csv`
- `responses_scored.csv`

Thematic Analysis By Question (Unsupervised)

- Discovers themes separately inside each question.
- Enable:
- `thematic_analysis.by_question.enabled: true`
- Key output:
- `question_theme_summary.csv`

Thematic Coding By Question (Supervised)

- Applies question-specific label sets with zero-shot classification.
- Enable:
- `thematic_analysis.by_question_zero_shot.enabled: true`
- Configure:
- `labels_by_question` in YAML.
- Key output:
- `question_label_summary.csv`

3) Strict Coding Mode (Recommended)

- Purpose: send ambiguous responses to manual review.

- Enable:
- `zero_shot.strict_coding.enabled: true` (global)
- `thematic_analysis.by_question_zero_shot.strict_coding.enabled: true` (by question)

Suggested Starting Thresholds
- `top1_min_score: 0.60`
- `min_margin_over_second: 0.12`
- `uncertain_label: "UNCERTAIN_REVIEW"`

Strict Mode Outputs
- `strict_review_queue_global.csv`
- `strict_review_queue_question.csv`

4) How To Read Results
1. Start with `run_report.md`.
2. Review `cluster_summary.csv` for major themes.
3. Validate themes with representative quotes.
4. If strict mode is on, manually review all `UNCERTAIN_REVIEW` rows.
5. Adjust labels and thresholds, then rerun.

5) Common Mistakes To Avoid
- Treating clusters as final truth without quote review.
- Using vague labels (example: "Other issue") too early.
- Setting strict thresholds too high on small datasets.
- Comparing themes across questions without using question-level mode.

6) Troubleshooting
- Error: column not found
- Check exact text in `input.text_column` and `input.question_column`.
- Too many uncertain rows
- Lower `top1_min_score` slightly (example: 0.60 -> 0.55).
- Themes look mixed
- Increase label quality or use question-level analysis.

7) Deliverable Checklist
- `run_report.md` shared with stakeholders.
- Theme table exported (`cluster_summary.csv` or `question_theme_summary.csv`).
- Uncertain queue reviewed and resolved.
- Final coding rules documented for repeatability.